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Gradient Descent and mathematics behind it.



It is an optimisation method that finds the minimum of an objective function by incrementally updating its parameters in its negative direction of the gradient of the function which is the direction of the steepest descent.

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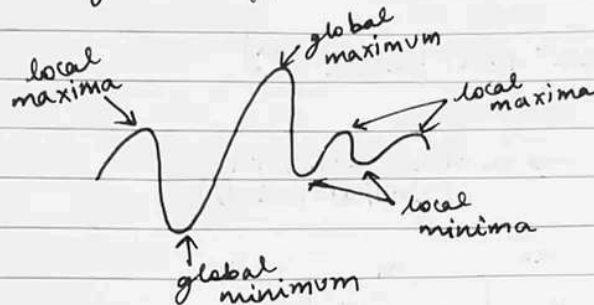
Go through the notes and the concept will be crystal clear!

classmate

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Gradient descent is a method for unconstrained mathematical optimization. It is a first order iterative algorithm for minimizing a different multivariate function.

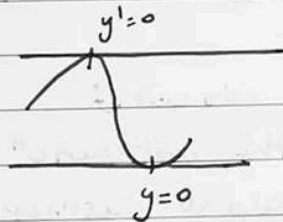


Minimum is lowest value in set & maximum is the highest value.
Global means it is true for entire set.
Local means it is true in some vicinity.

How do we find em? $\rightarrow$ Calculus.

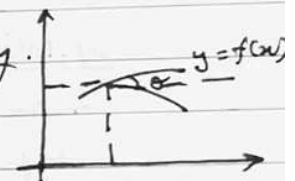
Derivatives give rate of change of x wrt y .

critical point is: $\tan \theta = 0$



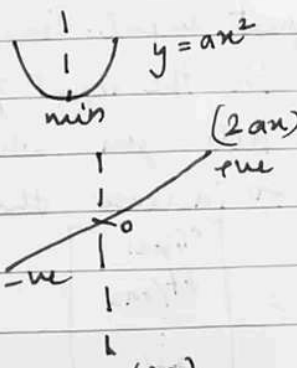
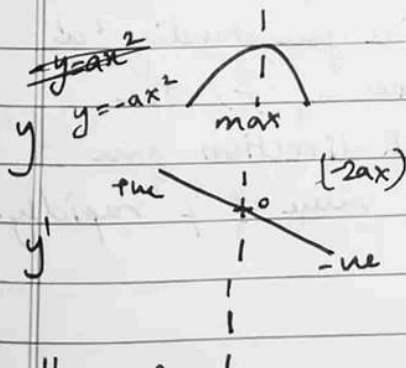
$\therefore \frac{dy}{dx} = 0 \Rightarrow$ The tangent is
||el to the
horizontal axis.

Regardless of local or
global points its
min $\frac{dy}{dx} = 0$.



$$\left. \frac{dy}{dx} \right|_{x=x_0} = \tan \theta$$

How to identify if critical point is local or global,
maximum or minimum?



- i) find critical points
- ii) derivative test
- iii) Max or min
- iv) local or global

f.

$$y = x^4 + x^3 - 6x^2$$

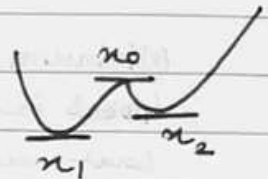
$$y' = 4x^3 + 3x^2 - 12x \quad (1^{st})$$

$$y'' = 12x^2 + 6x - 12 \quad (2^{nd})$$

$$y' = 0 \rightarrow x = 0, -2.15, 1.4$$

(critical points)

	x_0	x_1	x_2
x	0	-2.15	1.4
y''	-12	30.57	19.92
y	0	-16.3	-5.17
	local max	global min	local min



Optimizing is maxⁿ or minⁿ some function relative of some set having a range of choices. available in certain range. The funcⁿ allows comparison of diff choices which is to determine the 'best'.

Application: minimal concept, max profit, min error;

In DL optimization is to minimize the cost funcⁿ w.r.t the model parameters i.e. the weight matrices.

What is gradient?

∇f is a vector valued function and points in the direction of steepest ascent. Imagine you standing 'at' some point of in the input space of f , the gradient vector tells you which direction you should travel, to increase the value of 'f' rapidly.

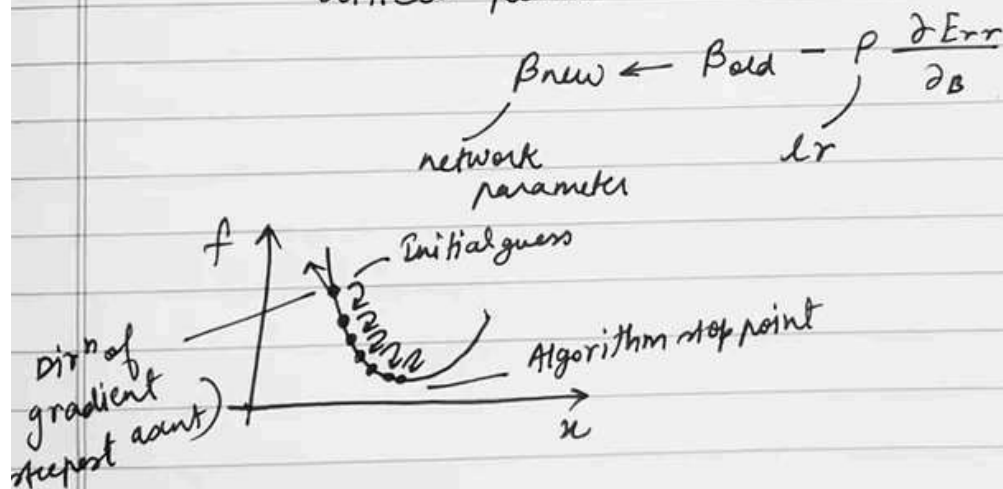
$$\nabla f = \begin{bmatrix} \partial f / \partial x_1 \\ \partial f / \partial x_2 \\ \vdots \\ \partial f / \partial x_n \end{bmatrix}$$

∴ Gradient Descent

This is a type of optimization technique where you move towards the minimum in an iterative manner.

Steps:

- i) Guess a point
- ii) Choose lr and it is also one of the most imp hyperparameters in DL
- iii) Calculate gradient of the function
- iv) move in opposite dirⁿ by subtracting this from initial guess. This is because we want to descend & gradient gives you steepest ascent direction.
- v) Repeat steps 3 & 4 for n times or until critical point.



∴ It is an optimization method that finds the minimum of an objective function by incrementally updating its parameters in its negative direction of the gradient of the function which is the direction of steepest descent.

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Written by Shetye Swara

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